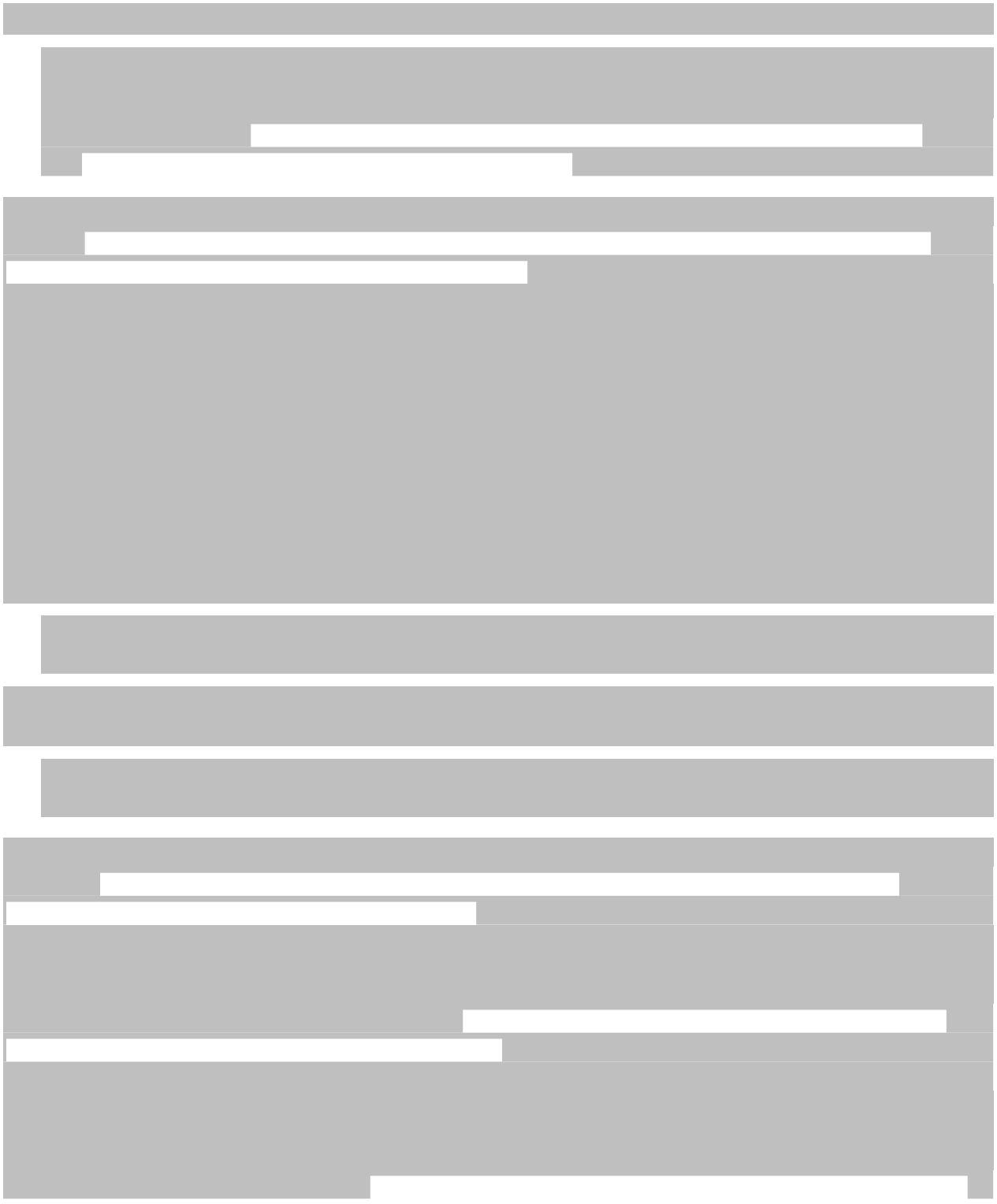
The FDA's Assessment:

FDA agrees with use of treatment-regimen estimand for evaluation of efficacy endpoints and for proposed labeling. However, there are multiple statistical issues with the Applicant's detailed implementation of the primary analyses.

FDA disagrees with the Applicant's imputation approaches for the primary analysis. For

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FDA agrees with the Applicant's multiplicity control strategy using graphical approach and conduct across all primary and key secondary endpoints, as prespecified in the SAP.

Note that all efficacy results reported in this review were based on the FDA's preferred multiple imputation (i.e., modified multiple imputation in the Applicant's analysis) and it applied to both ATTAIN-1 and ATTAIN-2. A total of 100 imputed datasets[3] were generated. In addition, all

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FDA Table 21. Summary of Observed/Missing Data at Week 72, ATTAIN-1

Parameter	Placebo N=949 n (%)	OFG 6 mg N=723 n (%)	OFG 12 mg N=725 n (%)	OFG 36 mg N=730 n (%)
Observed at Week 72	722 (76.1)	609 (84.2)	625 (86.2)	622 (85.2)
On treatment	652 (68.7)	561 (77.6)	560 (77.2)	549 (75.2)
Retrieved dropouts	70 (7.4)	48 (6.6)	65 (9.0)	73 (10.0)
Missing at Week 72	227 (23.9)	114 (15.8)	100 (13.8)	108 (14.8)
On treatment[1]	13 (1.4)	4 (0.6)	2 (0.3)	3 (0.4)
Off treatment[2]	214 (22.6)	110 (15.2)	98 (13.5)	105 (14.4)

Source: FDA analysis; adsl.xpt, advs.xpt
[1] Missing values at Week 72 were imputed based on missing at random.
[2] Missing values at Week 72 were imputed using a retrieved dropouts multiple imputation.
Cell content shows frequency and percentage relative to N in the parentheses.
Abbreviations: N=number of subjects randomized, n=number of subjects with specified characteristic, OFG=orforglipron

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FDA Table 23. Primary Analysis: Percentage Change in Body Weight at Week 72, ATTAIN-1

Parameter	Placebo N=949	OFG 6 mg N=723	OFG 12 mg N=725	OFG 36 mg N=730
Number of missing at baseline, n (%)	0	0	0	0
Missing at Week 72, n (%)	227 (23.9)	114 (15.8)	100 (13.8)	108 (14.8)
Baseline body weight (kg), mean (SD)	103.9 (22.0)	103.2 (21.7)	102.2 (21.6)	103.1 (23.2)
Estimate (%), LSMean (SE)[1]	-2.1 (0.3)	-7.4 (0.3)	-8.3 (0.3)	-11.2 (0.4)
Difference from placebo, (95% CI)		-5.3 (-6.1, -4.5)	-6.2 (-7.1, -5.3)	-9.0 (-10.0, -8.1)
p-value		<0.001	<0.001	<0.001

Source: FDA analysis; adsl.xpt, advs.xpt
[1] Analysis of covariance model for endpoint included treatment, region, strata, baseline and treatment by strata and treatment by baseline interaction terms; imputation for missing values at Week 72 was based on modified multiple imputations.
Abbreviations: CI=confidence interval, LSMean=least squares mean, N=number of subjects randomized, OFG=orforglipron, SD=standard deviation, SE=standard error

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Reference ID: 5773578

FDA Table 24. Key Secondary Endpoints: Body Weight Loss and Waist Circumference at Week 72, ATTAIN-1

Parameter	Placebo N=949	OFG 6 mg N=723	OFG 12 mg N=725	OFG 36 mg N=730
Missing at baseline, n (%)	0	0	0	0
Missing at Week 72, n (%)	227 (23.9)	114 (15.8)	100 (13.8)	108 (14.8)
≥5% body weight loss				
Number of responders at Week 72, n (%)	257 (27.1)	431 (59.6)	458 (63.2)	522 (71.5)
Risk (%), mean (SE)[1]	26.8 (1.6)	59.6 (1.9)	63.1 (1.9)	71.5 (1.8)
Difference from placebo, (95% CI)		32.8 (27.9, 37.6)	36.3 (31.5, 41.1)	44.7 (40.0, 49.3)
p-value		<0.001	<0.001	<0.001
≥10% body weight loss				
Number of responders at Week 72, n (%)	125 (13.2)	235 (32.5)	289 (39.9)	397 (54.4)
Risk (%), mean (SE)[1]	13.0 (1.3)	32.5 (1.8)	39.8 (1.9)	54.5 (1.9)
Difference from placebo, (95% CI)		19.4 (15.2, 23.7)	26.8 (22.4, 31.2)	41.4 (37.0, 45.9)
p-value		<0.001	<0.001	<0.001
≥15% body weight loss				
Number of responders at Week 72, n (%)	58 (6.1)	104 (14.4)	146 (20.1)	261 (35.8)
Risk (%), mean (SE)[1]	6.0 (0.9)	14.3 (1.3)	20.1 (1.5)	35.9 (1.8)
Difference from placebo, (95% CI)		8.3 (5.2, 11.4)	14.1 (10.7, 17.6))	29.8 (25.9, 33.7)
p-value		<0.001	<0.001	<0.001
≥20% body weight loss				
Number of responders at Week 72, n (%)	28 (3.0)	43 (5.9)	65 (9.0)	134 (18.4)
Risk (%), mean (SE)[1]	2.9 (0.6)	5.9 (0.9)	8.9 (1.1)	18.4 (1.5)
Difference from placebo, (95% CI)		3.0 (0.9, 5.2)	6.0 (3.6, 8.4)	15.5 (12.4, 18.6)
p-value[3]		Not applicable	<0.001	<0.001

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Parameter	Placebo N=949	OFG 6 mg N=723	OFG 12 mg N=725	OFG 36 mg N=730
Waist circumference (cm): Change from baseline at Week 72				
Baseline, mean (SD)	112.8 (14.5)	112.2 (14.1)	112.0 (14.3)	112.4 (15.3)
Estimate, LSMean (SE)[2]	-3.1 (0.3)	-7.0 (0.3)	-8.2 (0.3)	-10.0 (0.4)
Difference from placebo, (95% CI)		-3.9 (-4.7, -3.1)	-5.1 (-5.9, -4.3)	-6.9 (-7.8, -6.0)
p-value		<0.001	<0.001	<0.001

Source: FDA analysis; adsl.xpt, advs.xpt
[1] Logistic regression model for endpoint included treatment, region, strata, baseline, and treatment by strata and treatment by baseline interaction terms.
[2] Analysis of covariance model for endpoint included treatment, region, strata, baseline, and treatment by strata and treatment by baseline interaction terms; imputation for missing values at Week 72 was based on modified multiple imputations.
[3] p-values are reported only for parameters controlled for multiplicity.
Abbreviations: CI=confidence interval, LSMean=least squares mean, N=number of subjects randomized, n=number of subjects with the specified, OFG=orforglipron, SD=standard deviation, SE=standard error

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The FDA's Assessment:

Refer to the statistical analysis plan and amendments in Section 8.1.1.

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FDA Table 28. Summary of Observed/Missing Data at Week 72, ATTAIN-2

Parameter	Placebo N=630 n (%)	OFG 6 mg N=329 n (%)	OFG 12 mg N=332 n (%)	OFG 36 mg N=322 n (%)
Observed at Week 72	550 (87.3)	286 (86.9)	298 (89.8)	299 (92.9)
On treatment	503 (79.8)	266 (80.9)	257 (77.4)	256 (79.5)
Retrieved dropouts	47 (7.5)	20 (6.1)	41 (12.3)	43 (13.4)
Missing at Week 72	80 (12.7)	43 (13.1)	34 (10.2)	23 (7.1)
On treatment[1]	1 (0.2)	0	1 (0.3)	0
Off treatment[2]	79 (12.5)	43 (13.1)	33 (9.9)	23 (7.1)

Source: FDA analysis; adsl.xpt, advs.xpt
[1] Missing values at Week 72 were imputed based on missing at random.
[2] Missing values at Week 72 were imputed using a retrieved dropouts multiple imputation.
Cell content shows frequency and percentage relative to N in the parentheses.
Abbreviations: N=number of subjects randomized, n=number of subjects with specified characteristic, OFG=orforglipron

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FDA Table 30. Primary Analysis: Percentage Change in Body Weight at Week 72, ATTAIN-2

Parameter	Placebo N=630	OFG 6 mg N=329	OFG 12 mg N=332	OFG 36 mg N=322
Missing at Week 72, n (%)	80 (12.7)	43 (13.1)	34 (10.2)	23 (7.1)
Baseline body weight (kg), mean (SD)	101.2 (22.6)	102.3 (22.7)	102.7 (21.3)	99.8 (23.0)
Estimate (%), LSMean (SE)[1]	-2.5 (0.3)	-5.1 (0.4)	-7.0 (0.4)	-9.6 (0.5)
Difference from placebo, (95% CI)		-2.6 (-3.5, -1.6)	-4.5 (-5.4, -3.6)	-7.1 (-8.1, -6.1)
p-value		<0.001	<0.001	<0.001

Source: FDA analysis; adsl.xpt, advs.xpt
[1] Analysis of covariance model for endpoint included treatment, region, strata, baseline, and treatment by strata and treatment by baseline interaction terms; imputation for missing values at Week 72 was based on modified multiple imputation.
Abbreviations: CI=confidence interval, LSMean=least squares mean, N=number of subjects randomized, n=number of subjects with the specified characteristic, OFG=orforglipron, SD=standard deviation, SE=standard error

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FDA Table 31. Key Secondary Endpoints: Body Weight Loss and Waist Circumference at Week 72, ATTAIN-2

Endpoint	Placebo N=630	OFG 6 mg N=329	OFG 12 mg N=332	OFG 36 mg N=322
Body weight loss				
Missing at baseline, n (%)	0	0	0	0
Missing at Week 72, n (%)	80 (12.7)	43 (13.1)	34 (10.2)	23 (7.1)
≥5% Body weight loss				
Number of responders at Week 72, n (%)	169 (26.8)	157 (47.7)	183 (55.1)	217 (67.4)
Risk (%), mean (SE)[1]	26.8 (1.9)	47.4 (2.8)	54.7 (2.8)	67.0 (2.6)
Difference from placebo, (95% CI)		20.6 (13.9, 27.3)	27.9 (21.3, 34.5)	40.2 (33.8, 46.6)
p-value		<0.001	<0.001	<0.001
≥10% Body weight loss				
Number of responders at Week 72, n (%)	58 (9.2)	74 (22.5)	103 (31.0)	147 (45.7)
Risk (%), mean (SE)[1]	9.2 (1.3)	22.5 (2.4)	31.1 (2.6)	45.6 (2.7)
Difference from placebo, (95% CI)		13.3 (8.1, 18.5)	21.9 (16.3, 27.5)	36.4 (30.5, 42.3)
p-value		<0.001	<0.001	<0.001
≥15% Body weight loss				
Number of responders at Week 72, n (%)	19 (3.0)	22 (6.7)	47 (14.2)	85 (26.4)
Risk (%), mean (SE)[1]	3.1 (0.7)	6.7 (1.5)	14.4 (1.9)	25.9 (2.4)
Difference from placebo, (95% CI)		3.6 (0.4, 6.8)	11.3 (7.3, 15.3)	22.9 (17.9, 27.8)
p-value[3]		Not applicable	<0.001	<0.001
≥20% Body weight loss[4]				
Number of responders at Week 72, n (%)	5 (0.8)	8 (2.4)	15 (4.5)	35 (10.9)
Risk (%), mean (SE)[1]	0.7 (0.4)	2.6 (0.9)	4.4 (1.1)	10.8 (1.7)
Difference from placebo, (95% CI)		1.8 (-0.2, 3.8)	3.7 (1.4, 6.0)	10.1 (6.6, 13.5)
p-value[3]		Not applicable	Not applicable	Not applicable

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Endpoint	Placebo N=630	OFG 6 mg N=329	OFG 12 mg N=332	OFG 36 mg N=322
Waist circumference (cm): Change from baseline at Week 72				
Missing at baseline, n (%)	0	0	0	0
Missing at Week 72, n (%)	81 (12.9)	44 (13.4)	34 (10.2)	23 (7.1)
Baseline, mean (SD)	115.0 (14.6)	116.8 (15.1)	116.2 (13.4)	114.7 (15.1)
Estimate, LSMean (SE)[2]	-2.8 (0.3)	-5.4 (0.4)	-6.3 (0.4)	-8.3 (0.4)
Difference from placebo, (95% CI)		-2.6 (-3.5, -1.6)	-3.5 (-4.5, -2.6)	-5.5 (-6.5, -4.6)
p-value[3]		Not applicable	Not applicable	<0.001

Source: FDA analysis, adsl.xpt, advs.xpt
[1] Logistic regression model for endpoint included treatment, region, strata, baseline, and treatment by strata and treatment by baseline interaction terms.
[2] Analysis of covariance model for endpoint included treatment, region, strata, baseline, and treatment by strata and treatment by baseline interaction terms; imputation for missing values at Week 72 was based on modified multiple imputation.
[3] p-values are reported only for parameters controlled for multiplicity.
[4] Not a key secondary endpoint
Abbreviations: CI=confidence interval, N=number of subjects randomized, OFG=orforglipron, SD=standard deviation, SE=standard error

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